Supply Chain & Inventory Analytics

Project Overview

This case study presents an end-to-end **Supply Chain & Inventory Analytics** project developed using **SQL, Excel, and Power BI**. The project focuses on analyzing order performance, inventory levels, warehouse operations, supplier efficiency, and delivery performance to generate actionable business insights.

The analysis is based on two interconnected datasets containing **order transactions and inventory records**, enabling a comprehensive view of supply chain operations. The project follows the complete analytics workflow, including data cleaning, SQL analysis, dashboard development, and business insight generation to support operational decision-making.

Analysis Objective

The objective of this project is to analyze supply chain and inventory data to improve operational efficiency, monitor warehouse performance, and support better inventory management through interactive dashboards and business analysis.

- Evaluate warehouse performance using revenue and order volume.
- Identify products that require immediate replenishment.
- Measure delivery efficiency across different cities.
- Analyze supplier performance and inventory availability.
- Build an interactive dashboard to support operational decision-making

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Dataset Description

This project uses **two related datasets** to analyze the complete supply chain process, from customer orders to inventory management.

Orders Table

The Orders table contains transactional data related to customer purchases and logistics operations.

- Key Information
- Order Details
- Product Information
- Warehouse Details
- Supplier ID
- Customer City
- Revenue
- Shipping Cost
- Delivery Days
- Order Status
- Payment Method

Inventory Table

The Inventory table provides information about product availability and warehouse stock.

Key Information

- Product ID
- Warehouse ID
- Supplier Name
- Stock Available
- Reorder Level
- Lead Time
- Unit Cost
- Inventory Value

Dataset Summary

- **Orders Table:** ~10,000+ records
- **Inventory Table:** ~800+ records
- **Relationship:** Product_ID
- **Tools Used:** SQL, Excel, Power BI

Tools & Technologies Used

- **Database:** MySQL – Data storage, relational database management, and SQL analysis.
- **SQL:** Joins, Aggregate Functions, Window Functions, CTEs, Subqueries, and Business Queries.
- **Data Handling:** CSV files, Data Cleaning, Missing Value Handling, Duplicate Removal, and Data Validation.
- **Google Sheets:** Data Cleaning, Formulas, Pivot Tables, Charts, and KPI Reporting.
- **Power BI:** Power Query, Data Modeling, DAX Measures, Interactive Dashboards, and Data Visualization.

Analysis Performed

The supply chain data was analyzed using SQL, Excel, and Power BI to evaluate operational performance, inventory efficiency, and warehouse productivity. The analysis focused on identifying trends, measuring KPIs, and generating insights to support business decisions.

1. Warehouse Performance Analysis

- Analyzed revenue generated by each warehouse.
- Ranked warehouses based on overall sales performance.

2. Inventory Analysis

- Identified products below the reorder level.
- Evaluated inventory value across different warehouses.

3. Order & Delivery Analysis

- Measured average delivery time by customer city.
- Analyzed delayed orders and overall delivery performance.

4. Product & Category Analysis

- Identified top-performing product categories.
- Analyzed products generating the highest revenue.

5. Supplier Analysis

- Compared supplier performance based on shipping costs.
- Evaluated supplier contribution to inventory availability.

6. Dashboard & KPI Analysis

- Developed interactive dashboards using Excel and Power BI.
- Monitored KPIs such as Total Revenue, Total Orders, Inventory Value, Average Delivery Days, Delayed Orders, and Products Below Reorder Level.

Business Insights

Key Business Insights

- Warehouse **W03** generated the highest revenue, indicating strong operational performance.
- Several products were identified **below the reorder level**, highlighting the need for timely inventory replenishment.
- A few customer cities experienced **higher average delivery times**, suggesting opportunities to optimize logistics.
- Inventory value was concentrated in a small number of warehouses, indicating uneven stock distribution.
- Delayed deliveries had a measurable impact on overall supply chain efficiency and customer service.

Recommendations

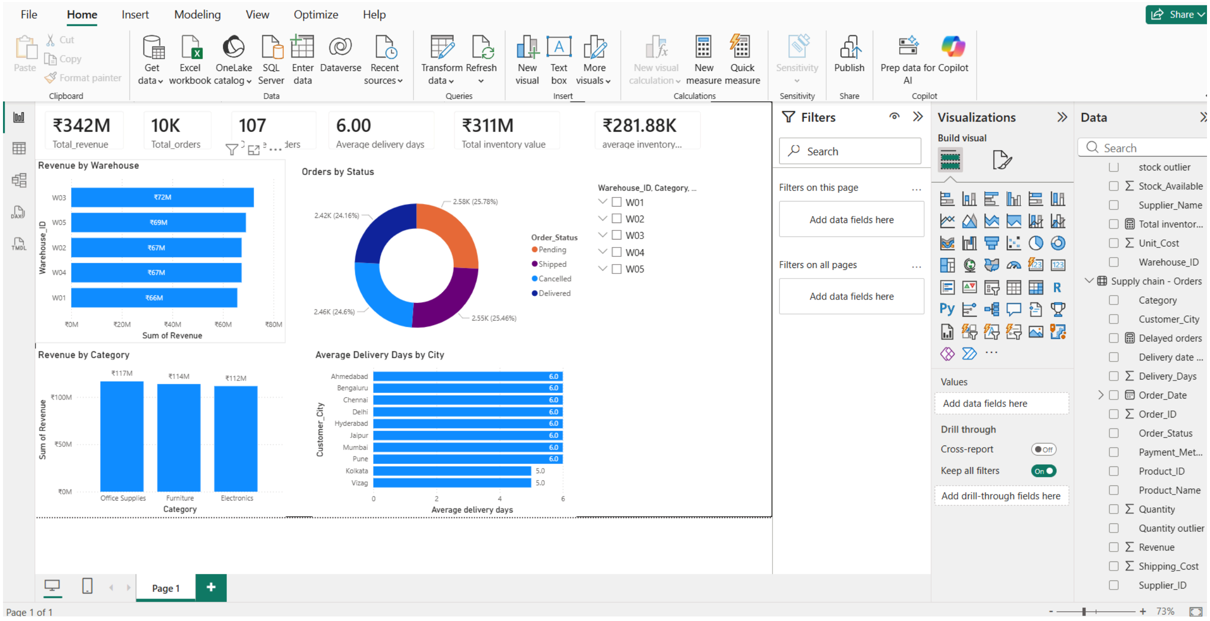
- Prioritize inventory replenishment for products below the reorder level to prevent stockouts.
- Optimize warehouse stock allocation to balance inventory across locations.
- Improve delivery planning for cities with consistently higher delivery times.
- Monitor supplier performance regularly to reduce lead times and improve operational efficiency.
- Track key supply chain KPIs through interactive dashboards to support proactive decision-making.

Resume-Ready Project Summary

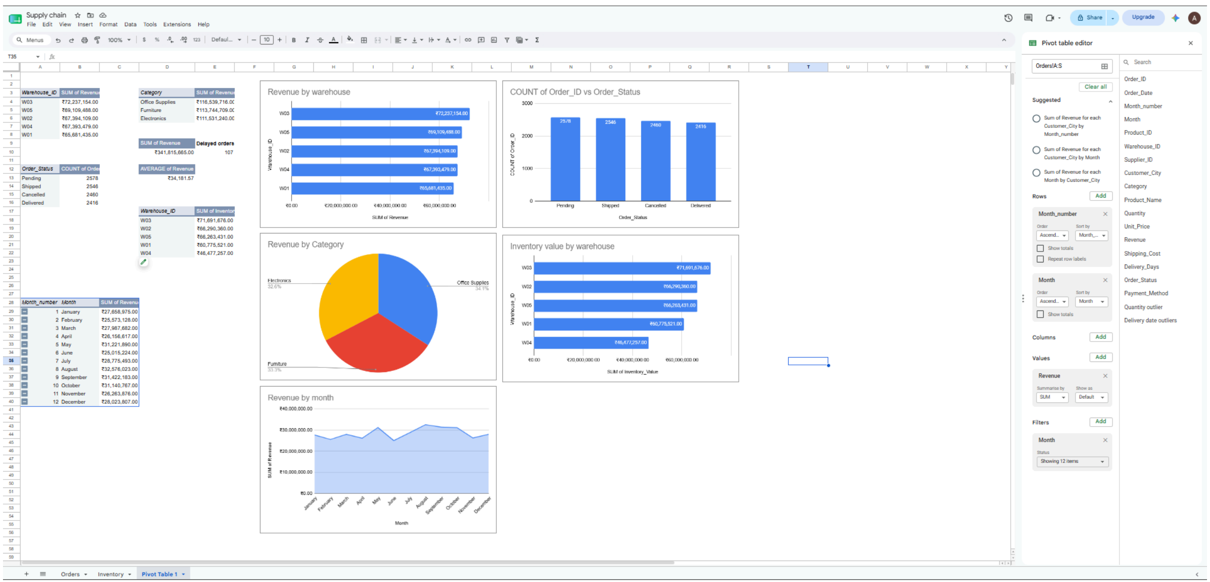
Supply Chain & Inventory Analytics Project | SQL, Excel, Power BI

- Examined supply chain operations by analyzing order fulfillment, warehouse activities, inventory availability, and supplier performance across two connected datasets.
- Measured warehouse efficiency, product movement, and inventory levels to identify operational bottlenecks and improve stock management.
- Performed business analysis using SQL to evaluate warehouse revenue, delivery performance, reorder requirements, supplier efficiency, and inventory valuation.
- Designed interactive dashboards in Power BI and Google Sheets to monitor operational KPIs, warehouse performance, and monthly supply chain trends.
- Identified inventory risks, delivery delays, and stock distribution patterns to support faster operational decision-making.
- Proposed data-driven recommendations to improve inventory planning, optimize warehouse utilization, and enhance overall supply chain efficiency.

Power BI:



Sheets:



Conclusion

The project identified key operational trends, highlighted inventory risks, and provided recommendations to improve supply chain efficiency. The interactive dashboards and business analysis enable stakeholders to monitor critical KPIs, make informed decisions, and optimize overall logistics performance.